PROJECT REPORT / SEMICON

Executive Summary

A Python and Tableau review of sales value, units, markets, customers, products and monthly performance

Overview of the Report

This project turns Semicon's transaction data and four supporting tables into a cleaned, related reporting dataset and a Tableau workbook. The two supplied dashboard views provide an Executive Overview of sales distribution and a Yearly Growth view of monthly trends. The workbook also contains supporting story points on customer concentration, market performance, product performance and business actions.

The source transactions cover October 2017 to June 2020. Python cleaning standardizes text and dates, removes duplicates and invalid or non-positive transactions, converts USD using a fixed 96 INR/USD rate, and checks transaction keys against the dimension tables.

Executive-Level Summary

The cleaned dataset retains 49,144 transactions and records 1.026 million units across 15 active markets and 38 customers. Summing the converted sales amount directly gives INR 308.58 million for the period. Delhi NCR leads the market sales-value view, while Mumbai leads unit volume. The top customer shown in the dashboard is Electricalsara Stores, and the top product by the dashboard's sales-amount ranking is the Semicon Vortex V9-1200K Unlocked Desktop CPU.

The monthly trend view peaks in January 2018 at about INR 17.4 million and reaches about INR 4.5 million in June 2020. Full-year converted sales amount totals are INR 123.35 million in 2018 and INR 110.04 million in 2019, a decline of about 10.8%. The 2017 and 2020 totals cover partial years and should not be compared with full-year totals as if they represented equivalent periods.

Key Facts from the Project

Measure	Value / interpretation
Clean transactions	49,144 rows retained from 50,000 source rows
Reporting period	October 2017 to June 2020
Unit volume	1,026,332 units
Active markets / customers	15 markets / 38 customers
Converted sales amount sum	INR 308.58M across cleaned transactions
Top market by sales amount	Delhi NCR - INR 110.73M
Top market by units	Mumbai - 191,125 units
Top customer by sales amount	Electricalsara Stores - INR 73.90M
Top product by sales amount	Semicon Vortex V9-1200K - INR 23.58M

Detailed Findings

1. Market Performance

Delhi NCR is the largest market in the sales-amount view at INR 110.73M, followed by Mumbai at INR 58.09M, Ahmedabad at INR 49.62M and Nagpur at INR 37.85M. Mumbai leads units sold with 191.1K, slightly ahead of Delhi NCR at 183.2K. Comparing value with volume can help identify markets with different product or transaction mixes, but the dashboard does not provide margin or average selling-price analysis.

2. Customer Concentration

Electricalsara Stores is the top customer at INR 73.90M in the dashboard ranking, followed by Excel Stores at INR 24.46M and Info Stores at INR 19.67M. The leading account contributes roughly 24% of the cleaned converted sales-amount total. This concentration supports closer account monitoring and a review of the sales mix behind the top account.

3. Product Contribution

The top product in the sales-amount ranking is the Semicon Vortex V9-1200K Unlocked Desktop CPU at INR 23.58M. The next two shown are Semicon Titan DDR4 8GB 3600MHz Desktop RAM at INR 16.26M and Semicon Vortex V9-1200 Desktop CPU at INR 15.60M. CPU and memory products feature prominently among the leading products.

4. Monthly and Annual Pattern

The monthly line chart shows its highest point in January 2018 at about INR 17.4M, followed by lower and uneven monthly results. A later high occurs in July 2019 at about INR 11.9M, while June 2020 is about INR 4.5M. Comparing complete years, converted sales amount falls from INR 123.35M in 2018 to INR 110.04M in 2019. The chart describes changes over time but does not explain their causes.

5. Coverage and Model

The Tableau model brings together transaction rows and customer, date, market and product dimensions. The dashboard reports 15 active markets and 38 selling customers. The market dimension contains 17 entries, so two listed markets have no transaction activity in the cleaned dataset. This distinction should stay clear when reporting market availability.

Important Metric Reconciliation

The workbook's headline "Revenue" KPI is calculated as converted sales amount multiplied by quantity, producing approximately INR 93,022.04M. The market, customer and product rankings and the monthly trend use converted sales amount summed directly, which totals INR 308.58M in the cleaned data. These are different calculation bases. Confirm whether source sales_amount is a transaction amount or a per-unit price, then align the KPI and all charts before using the headline as revenue or comparing it with the breakdowns.

Recommendations and Notes

Recommended Actions

- Reconcile the revenue definition first. Confirm the source meaning of sales_amount, then use one calculation and unit for every revenue KPI, trend and ranking. Add a reconciliation check to the workbook.
- Protect and review the largest accounts, beginning with Electricalsara Stores, and examine customer-level contribution over time to monitor concentration.
- Compare market sales value and unit volume together. Prioritize follow-up in leading markets such as Delhi NCR and Mumbai while reviewing lower-activity markets with local context.
- Review availability, pricing and promotion for the leading CPU and RAM products. The sales data identifies leaders but does not show inventory, margin or stock-out constraints.
- Investigate the 2018-2019 decline and uneven monthly changes with additional business context. Do not attribute the pattern to a cause based on this dashboard alone.
- Label partial years and the 96 INR/USD fixed conversion assumption directly in the report, and refresh the exchange rate when the analysis is updated.

Method and Interpretation

Topic	Definition used in this summary
Data cleaning	Notebook output has 49,144 cleaned transaction rows after deduplication and exclusion of invalid dates and non-positive sales records.
Sales amount	INR 308.58M is the sum of sales_amount_inr across cleaned transactions; it is the basis used for market/customer/product rankings and monthly trend.
Headline KPI	The Tableau field total_revenue multiplies sales_amount_inr by sales_qty. Its resulting INR 93,022.04M figure is kept separate from the direct amount sum until its intended meaning is confirmed.
Currency	Two USD transactions are converted at the notebook's fixed 96 INR per USD rate; this is an assumption, not a historical-rate conversion.
Period completeness	2017 covers October-December and 2020 covers January-June. 2018 and 2019 cover full calendar years.
Coverage	The cleaned transactions contain 15 of the 17 market dimension entries and all 38 customer dimension entries.

Conclusion

The project establishes a useful sales reporting foundation: cleaned transactions, linked product/customer/market/date dimensions, and dashboards that expose market, customer, product and time patterns. Its strongest immediate improvement is to reconcile and label the sales-value calculation. Once that is resolved, the dashboard can give leadership a consistent basis for monitoring sales performance and choosing where to investigate further.